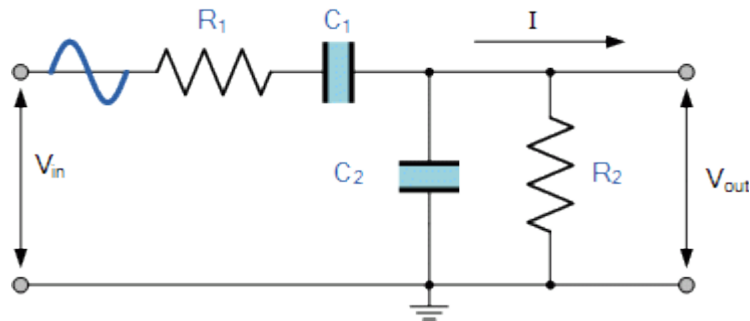




Passive Band Pass Filter

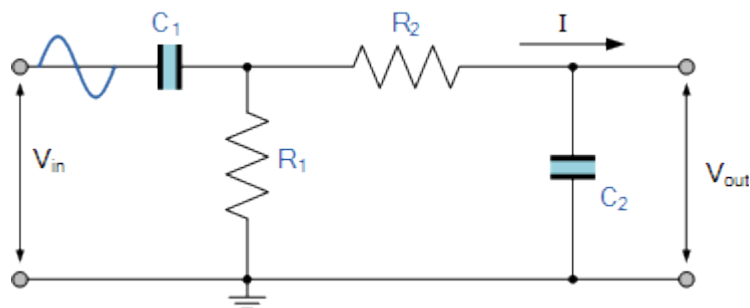
Passive Band Pass Filters can be made by connecting together a low pass filter with a high pass filter

Band Pass Filters can be used to isolate or filter out certain frequencies that lie within a particular band or range of frequencies. The cut-off frequency or f_c point in a simple RC passive filter can be accurately controlled using just a single resistor in series with a non-polarized capacitor, and depending upon which way around they are connected, we have seen that either a Low Pass or a High Pass filter is obtained.

One simple use for these types of passive filters is in audio amplifier applications or circuits such as in loudspeaker crossover filters or pre-amplifier tone controls. Sometimes it is necessary to only pass a certain range of frequencies that do not begin at 0Hz, (DC) or end at some upper high frequency point but are within a certain range or band of frequencies, either narrow or wide.

By connecting or “cascading” together a single **Low Pass Filter** circuit with a **High Pass Filter** circuit, we can produce another type of passive RC filter that passes a selected range or “band” of frequencies that can be either narrow or wide while attenuating all those outside of this range. This new type of passive filter arrangement produces a frequency selective filter known commonly as a **Band Pass Filter** or **BPF** for short.

Band Pass Filter Circuit



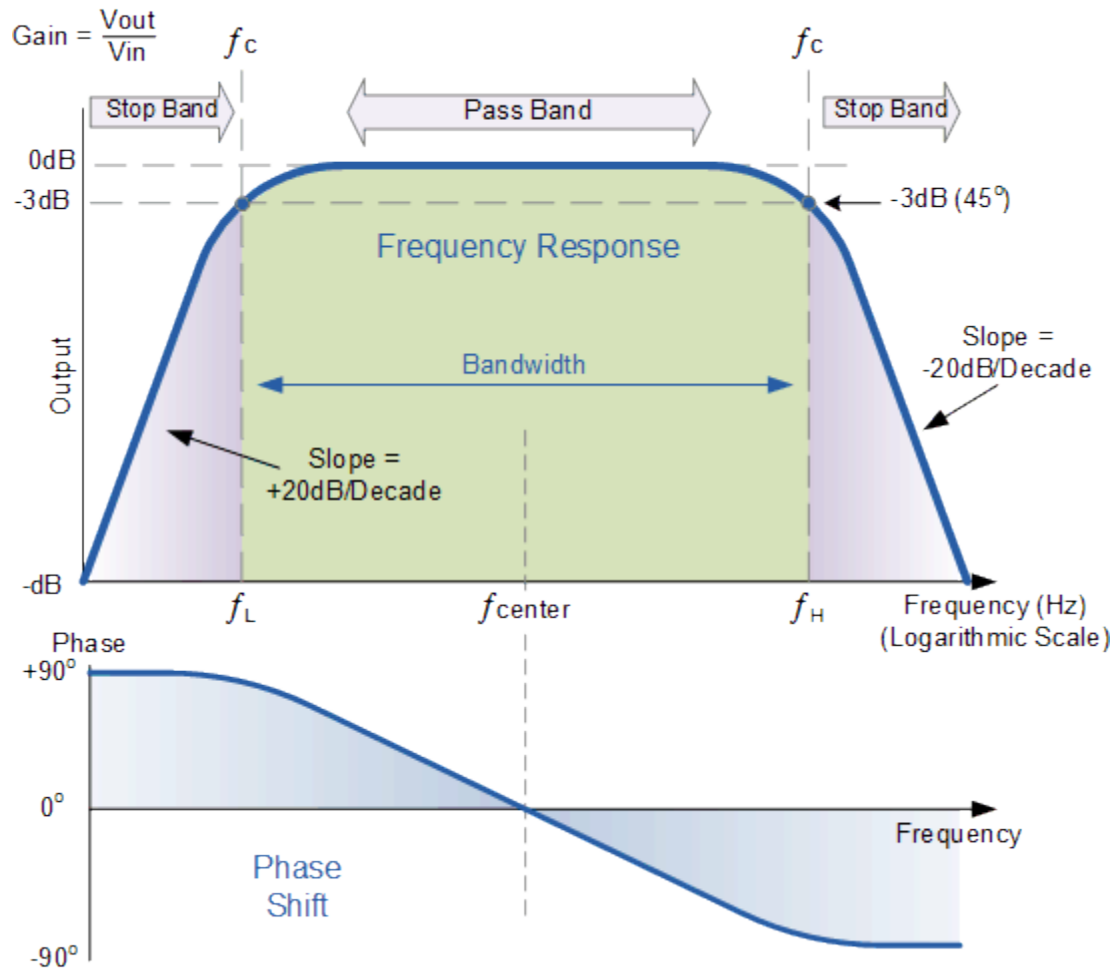
Unlike the low pass filter which only pass signals of a low frequency range or the high pass filter which pass signals of a higher frequency range, a **Band Pass Filters** passes signals within a certain “band” or “spread” of frequencies without distorting the input signal or introducing extra noise. This band of frequencies can be any width and is commonly known as the filters **Bandwidth**.

Bandwidth is commonly defined as the frequency range that exists between two specified frequency cut-off points (f_c), that are 3dB below the maximum centre or resonant peak while attenuating or weakening the others outside of these two points.

Then for widely spread frequencies, we can simply define the term “bandwidth”, BW as being the difference between the lower cut-off frequency (f_{LOWER}) and the higher cut-off frequency (f_{HIGHER}) points. In other words, $BW = f_H - f_L$. Clearly for a pass band filter to function correctly, the cut-off frequency of the low pass filter must be higher than the cut-off frequency for the high pass filter.

The “ideal” **Band Pass Filter** can also be used to isolate or filter out certain frequencies that lie within a particular band of frequencies, for example, noise cancellation. Band pass filters are known generally as second-order filters, (two-pole) because they have “two” reactive component, the capacitors, within their circuit design. One capacitor in the low pass circuit and another capacitor in the high pass circuit.

Frequency Response of a 2nd Order Band Pass Filter



The **Bode Plot** or frequency response curve above shows the characteristics of the band pass filter. Here the signal is attenuated at low frequencies with the output increasing at a slope of +20dB/Decade (6dB/Octave) until the frequency reaches the “lower cut-off” point f_L . At this frequency the output voltage is again $1/\sqrt{2} = 70.7\%$ of the input signal value or **-3dB** ($20 \cdot \log(V_{OUT}/V_{IN})$) of the input.

The output continues at maximum gain until it reaches the “upper cut-off” point f_H where the output decreases at a rate of -20dB/Decade (6dB/Octave) attenuating any high frequency signals. The point of maximum output gain is generally the geometric mean of the two -3dB value between the lower and upper cut-off points and is called the “Centre Frequency” or “Resonant Peak” value f_r . This geometric mean value is calculated as being $f_r^2 = f_{(UPPER)} \times f_{(LOWER)}$.

A band pass filter is regarded as a second-order (two-pole) type filter because it has “two” reactive components within its circuit structure, then the phase angle will be twice that of the previously seen first-order filters, ie, **180°**. The phase angle of the output signal **LEADS** that of

the input by **+90°** up to the centre or resonant frequency, f_r point where it becomes “zero” degrees (0°) or “in-phase” and then changes to **LAG** the input by **-90°** as the output frequency increases.

The upper and lower cut-off frequency points for a band pass filter can be found using the same formula as that for both the low and high pass filters, For example.

$$f_c = \frac{1}{2\pi RC} \text{ Hz}$$

Then clearly, the width of the pass band of the filter can be controlled by the positioning of the two cut-off frequency points of the two filters.

Band Pass Filter Example No1.

A second-order **band pass filter** is to be constructed using RC components that will only allow a range of frequencies to pass above 1kHz (1,000Hz) and below 30kHz (30,000Hz). Assuming that both the resistors have values of 10kΩ, calculate the values of the two capacitors required.

The High Pass Filter Stage

The value of the capacitor C1 required to give a cut-off frequency f_L of 1kHz with a resistor value of 10kΩ is calculated as:

$$C_1 = \frac{1}{2\pi f_L R} = \frac{1}{2\pi \times 1,000 \times 10,000} = 15.9 \text{ nF}$$

Then, the values of R1 and C1 required for the high pass stage to give a cut-off frequency of 1.0kHz are: R1 = 10kΩ and to the nearest preferred value, C1 = 15nF.

The Low Pass Filter Stage

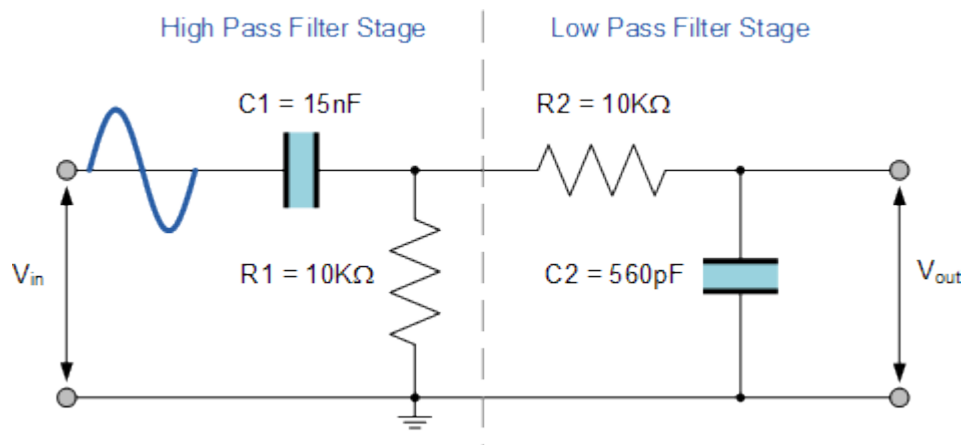
The value of the capacitor C2 required to give a cut-off frequency f_H of 30kHz with a resistor value of 10kΩ is calculated as:

$$C_2 = \frac{1}{2\pi f_H R} = \frac{1}{2\pi \times 30,000 \times 10,000} = 530 \text{ pF}$$

Then, the values of R2 and C2 required for the low pass stage to give a cut-off frequency of 30kHz are, $R = 10\text{k}\Omega$ and $C = 530\text{pF}$. However, the nearest preferred value of the calculated capacitor value of 530pF is 560pF, so this is used instead.

With the values of both the resistances R1 and R2 given as $10\text{k}\Omega$, and the two values of the capacitors C1 and C2 found for both the high pass and low pass filters as 15nF and 560pF respectively, then the circuit for our simple passive **Band Pass Filter** is given as.

Completed Band Pass Filter Circuit



Band Pass Filter Resonant Frequency

We can also calculate the “Resonant” or “Centre Frequency” (f_r) point of the band pass filter where the output gain is at its maximum or peak value. This peak value is not the arithmetic average of the upper and lower -3dB cut-off points as you might expect but is in fact the “geometric” or mean value. This geometric mean value is calculated as being $f_r^2 = f_{C(\text{UPPER})} \times f_{C(\text{LOWER})}$ for example:

Centre Frequency Equation

$$f_r = \sqrt{f_L \times f_H}$$

Where, f_r is the resonant or centre frequency

- f_L is the lower -3dB cut-off frequency point
- f_H is the upper -3db cut-off frequency point

and in our simple example above, the calculated cut-off frequencies were found to be $f_L = 1,060$ Hz and $f_H = 28,420$ Hz using the filter values.

Then by substituting these values into the above equation gives a central resonant frequency of:

$$f_r = \sqrt{f_L \times f_H} = \sqrt{1,060 \times 28,420} = 5,48 \text{ kHz}$$

Band Pass Filter Summary

A simple passive **Band Pass Filter** can be made by cascading together a single **Low Pass Filter** with a **High Pass Filter**. The frequency range, in Hertz, between the lower and upper -3dB cut-off points of the RC combination is known as the filters “Bandwidth”.

The width or frequency range of the filters bandwidth can be very small and selective, or very wide and non-selective depending upon the values of R and C used.

The centre or resonant frequency point is the geometric mean of the lower and upper cut-off points. At this centre frequency the output signal is at its maximum and the phase shift of the output signal is the same as the input signal.

The amplitude of the output signal from a band pass filter or any passive RC filter for that matter, will always be less than that of the input signal. In other words a passive filter is also an attenuator giving a voltage gain of less than 1 (Unity). To provide an output signal with a voltage gain greater than unity, some form of amplification is required within the design of the circuit.

A **Passive Band Pass Filter** is classed as a second-order type filter because it has two reactive components within its design, the capacitors. It is made up from two single RC filter circuits that are each first-order filters themselves.

If more filters are cascaded together the resulting circuit will be known as an “nth-order” filter where the “n” stands for the number of individual reactive components and therefore poles within the filter circuit. For example, filters can be a 2nd-order, 4th-order, 10th-order, etc.

The higher the filters order the steeper will be the slope at n times -20dB/decade. However, a single capacitor value made by combining together two or more individual capacitors is still one capacitor.